

$$\dot{T} = \gamma_{rel} - Q$$

$$P_{rel} = u$$

$$Q = \alpha T + \beta T^2$$

$$\dot{T} = \gamma u - \alpha T - \beta T^2$$

$$0 = \gamma u - \alpha T - \beta T^2$$

$$\frac{\alpha \pm \sqrt{\alpha^2 + 4\gamma\beta}}{-2\beta}$$

$$J = [-\alpha - 2\beta T] \Big|_{T = \frac{\alpha \pm \sqrt{\alpha^2 + 4\gamma\beta}}{-2\beta}}$$

$$= \left[-\alpha - 2\beta \left[\frac{\alpha \pm \sqrt{\alpha^2 + 4\gamma\beta}}{-2\beta} \right] \right]$$

$$= -\alpha + \alpha \pm \sqrt{\alpha^2 + 4\gamma\beta}$$

$$= \pm \sqrt{\alpha^2 + 4\gamma\beta}$$

$$\lambda = \pm \sqrt{\alpha^2 + 4\gamma\beta}$$

se $\lambda = -\sqrt{\alpha^2 + 4\gamma\beta}$ è stabile (u non può essere negativa)

se $\lambda = +\sqrt{\alpha^2 + 4\gamma\beta}$ è instabile